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Arduino UNO R4 Minima Real-Time Clock

Learn how to access the real-time clock
(RTC) on the UNO R4 Minima.

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In this tutorial you will learn how
to access the real-time clock (RTC)
on an **Arduino UNO R4 Minima**
board. The RTC is embedded in
the UNO R4 Minima's
microcontroller (RA4M1).

Goals

The goals of this project are:

Set a start date of the RTC

Access the date / time from
the RTC in calendar format.

Access the time in Unix
format.

Hardware & Software Needed

Arduino IDE ([online](#) or
[offline](#))

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Real-Time Clock (RTC)

The RTC on the UNO R4 Minima can be accessed using the [RTC](#) library that is included in the [UNO R4 Board Package](#). This library allows you to set/get the time as well as using alarms to trigger interrupts.

There are many practical examples using an RTC, and the examples provided in this page will help you get started with it.

Set Time

```
RTCTime startTime(30,  
Month::JUNE, 2023, 13,  
37, 00,  
DayOfWeek::WEDNESDAY,  
SaveLight::SAVING_TIME_  
ACTIVE)
```

```
RTC.setTime(startTime)
```

To set the starting time for the RTC, you can create an `RTCTime` object. Here you can specify the day, month, year, hour, minute, second, and specify day of week as well as daylight saving mode.

Then to set the time, use the `setTime()` method.

Example:

```
1  #include "RTC.h"
2
3  void setup() {
4      Serial.begin(9600);
5
6      RTC.begin();
7
8      RTCTime startTime(30
9
10     RTC.setTime(startTime
11 }
12
13 void loop() {
14 }
```

Get Time

```
RTC.getTime(currentTime
)
```

To retrieve the time, we need to create a `RTCTime` object, and use the `getTime()` method to retrieve the current time.

This example sets & gets the time and stores it in an `RTCTime` object called `currentTime`.

```
1  #include "RTC.h"
2
3  void setup() {
4      Serial.begin(9600);
5
6      RTC.begin();
7
8      RTCTime startTime(30
9
10     RTC.setTime(startTime
11 }
12
13 void loop() {
14     RTCTime currentTime;
15
16     // Get current time from RTC
17     RTC.getTime(currentTime
18 }
```

Print Date & Time

The above examples show how to set & get the time and store it in an object. This data can be retrieved by a series of methods:

`getDayOfMonth()``getMonth()``getYear()``getHour()``getMinutes()``getSeconds()`

The example below prints out the date and time from the `currentTime` object.

```
1  #include "RTC.h"
2
3  void setup() {
4      Serial.begin(9600);
5
6      RTC.begin();
7
8      RTCTime startTime(3
9
10     RTC.setTime(startTi
11 }
12
13 void loop() {
14     RTCTime currentTime
15
16     // Get current time
17     RTC.getTime(current
18
19     // Print out date (
20     Serial.print(curren
21     Serial.print("/");
22     Serial.print(Month2
23     Serial.print("/");
24     Serial.print(curren
25     Serial.print(" - ")
26
27     // Print time (HH/M
28     Serial.print(curren
```

Unix

```
currentTime.getUnixTime()  
()
```

To retrieve the Unix timestamp, use the `getUnixTime()` method.

```
1  #include "RTC.h"  
2  
3  void setup() {  
4      Serial.begin(9600);  
5  
6      RTC.begin();  
7  
8      RTCTime startTime(30  
9  
10     RTC.setTime(startTime  
11 }  
12  
13 void loop() {  
14     RTCTime currentTime;  
15  
16     // Get current time  
17     RTC.getTime(currentT  
18  
19     //Unix timestamp  
20     Serial.print("Unix t  
21     Serial.println(curre  
22  
23     delay(1000);  
24 }
```

Periodic Interrupt

A periodic interrupt allows you to set a recurring callback.

To use this, you will need to initialize the periodic callback, using the

```
setPeriodicCallback()
```

method:

```
RTC.setPeriodicCallback  
(periodic_cbk,
```

You will also need to create a function that will be called:

```
void periodicCallback()
{ code to be executed }
```



Note the IRQ has a very fast execution time. Placing a lot of code is not a good practice, so in the example below we are only switching a single flag, `irqFlag`.

The example below blinks a light every 2 seconds:

```
1  #include "RTC.h"
2
3  volatile bool irqFlag
4  volatile bool ledStat
5
6  const int led = LED_B
7
8  void setup() {
9      pinMode(led, OUTPUT
10
11      Serial.begin(9600);
12
13      // Initialize the R
14      RTC.begin();
15
16      // RTC.setTime() mu
17      // what date and ti
18      RTCTime mytime(30,
19      RTC.setTime(mytime)
20
21      if (!RTC.setPeriodi
22          Serial.println("E
23      }
24  }
25
26  void loop(){
27      if(irqFlag){
28          Serial.println("T
```

The period can be specified using the following enumerations:

`N2_TIMES_EVERY_SEC``N4_TIMES_EVERY_SEC``N8_TIMES_EVERY_SEC``N16_TIMES_EVERY_SEC``N32_TIMES_EVERY_SEC``N64_TIMES_EVERY_SEC``N128_TIMES_EVERY_SEC``N256_TIMES_EVERY_SEC`

Alarm Callback

```
RTC.setAlarmCallback(alarm_cbk, alarmtime, am)
```

```
1 unsigned long previousTime = 0;
2 const long interval = 1000;
3 bool ledState = false;
4
5 // Include the RTC library
6 #include "RTC.h"
7
8 void setup() {
9   //initialize Serial
10  Serial.begin(9600);
11
12  //define LED as output pin
13  pinMode(LED_BUILTIN, OUTPUT);
14
15  // Initialize the RTC
16  RTC.begin();
17
18  // RTC.setTime() method sets the
19  // what date and time
20  RTCTime initialTime;
21  RTC.setTime(initialTime);
22
23  // Trigger the alarm
24  RTCTime alarmTime;
25  alarmTime.setSecond(10);
26
27  // Make sure to only trigger the
28  AlarmMatch matchTime;
```

Summary

This tutorial shows how to use the RTC on the UNO R4 Minima, such as setting a start time, setting an alarm, or obtaining time in calendar or unix format.

Read more about this board in the [Arduino UNO R4 Minima documentation](#).

Suggest changes

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